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A palaeomagnetic study of Devonian sandstone from Central Spitsbergen

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Summary. Samples from the Devonian red bed succession of Wood Bay Formation from Central Spitsbergen were thermally demagnetized. The vector-subtraction analytical method of separation of NRM components revealed stable magnetic directions ascribed to a Devonian palaeofield. The mean palaeomagnetic direction: $D = 58°$; $I = 12°$ yields the palaeomagnetic pole position: 12°N; 135°E, close to the Lower Devonian pole for Great Britain, indicating that Spitsbergen was as similarly related to continental Europe as the British Islands at that time.

Key words: Spitsbergen, Old Red Sandstone, detrital magnetization, tectonics

Introduction

To reveal the complexity of the crust of Western and Central Europe and its history in a plate tectonic context more palaeomagnetic data are needed, particularly from Spitsbergen. After the Caledonian activity and its rejuvenation during the Upper Devonian (Harland 1969; Birkenmajer 1981), Spitsbergen remained tectonically stable until the opening of North Atlantic and Arctic basins, which caused the West Spitsbergen orogeny along the western edge of the island. As the Variscan orogeny hardly affected south Spitsbergen, palaeomagnetic studies may provide data undisturbed by the Variscan activity. Devonian rocks in Spitsbergen occur in two N–S trending grabens exposed in the northern part of the island and in a smaller one in the south. The Old Red Sandstone molasse, of late orogenic origin, was deposited in these grabens. The oldest sediments are in the north of the island and the most widespread deposits belong to the Wood Bay Formation. This formation consists of continental red coloured sandstones with siltstones and shales of Siegenian–Emsian age. There are two main sources of clastics (Friend & Moody-Stuard 1972). The first, to the south and southeast, supplied material to the basin by river systems, whereas the western part of basin was supplied from the northwest side.

The Wood Bay Formation was sampled in Dicksonfjorden, central Spitsbergen, outside of the Tertiary Orogenic Belt (Fig. 1). The sampling localitites are situated at sea-level and

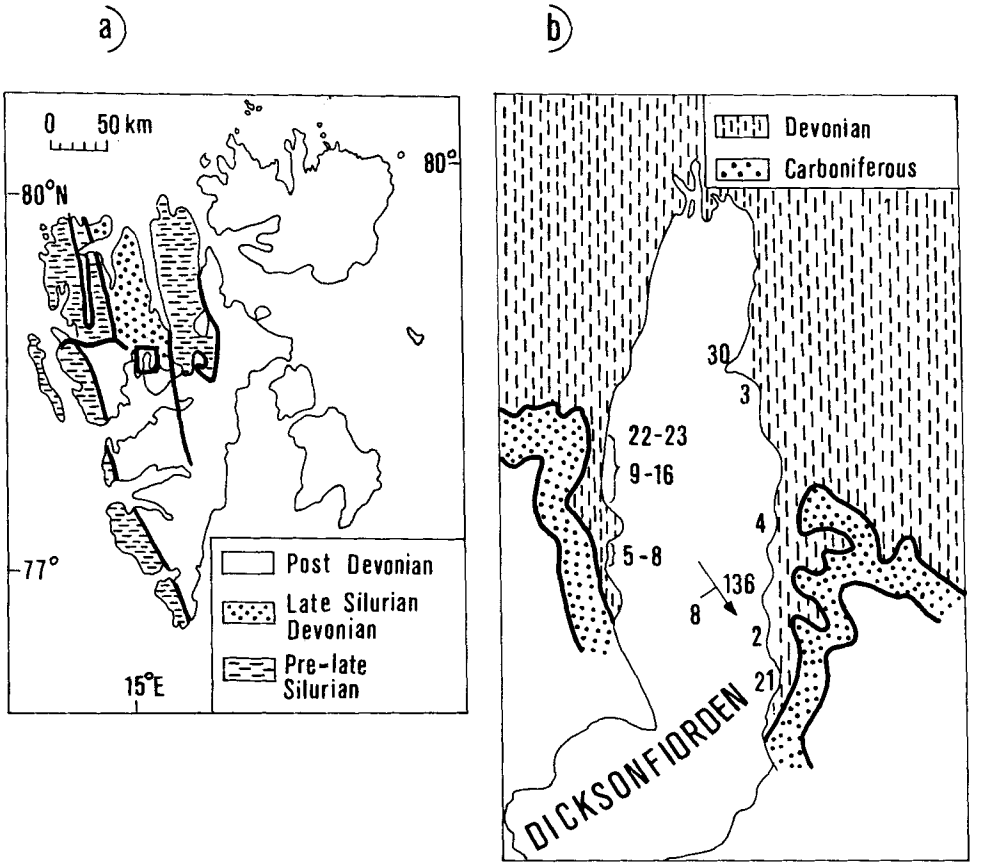


Figure 1. Maps of Spitsbergen and Dicksonfjorden with sampling sites shown by numbers, and regional bedding of the Wood Bay Formation red beds marked. (b) covers a 20 km × 34 km area.

cover the top 800 m of a red bed succession unconformably underlying middle Carboniferous limestones.

In the investigation of the Devonian red bed succession (Løvlie *et al.* 1984) palaeomagnetic and rock magnetic properties were compared. The samples were subjected to stepwise thermal demagnetization which revealed stable magnetic directions after heating to temperatures above 600°C. Rock magnetic properties (IRM acquisition, thermomagnetic analysis, changes in magnetic susceptibility after thermal demagnetization and rotational hysteresis) point to fine-grained hematite. Petrographic observations suggest a detrital origin of the hematite granules probably derived from a lateritic source area, and exposed to minor post-depositional oxidation. It suggests a depositional origin of magnetization carried by hematite grains, confirmed also by almost coinciding azimuthal distribution of remanent directions and axis of maximum susceptibility. Løvlie *et al.* (1984) did not give mean directions because of 'inclination error' which might be expected in rocks with depositional magnetization carried by detrital hematite (Løvlie & Torsvik 1984; Tauxe & Kent 1984). However, we came to the conclusion that the 'inclination error' should be cancelled out by randomizing effects of a high energy environment. Moreover, in high energy conditions a post-depositional RM recording the true direction of the ambient field should be present

(Barton, McElhinny & Edwards 1980; Payne & Verosub 1982; Tauxe & Kent 1984). So, we decided to re-examine the palaeomagnetic results and these are the subject of this paper.

The analysis of step demagnetization data

A total of 155 samples were subjected to progressive thermal demagnetization up to 700°C (Løvlie *et al.* 1984). Progressive alternating field demagnetization of pilot samples showed that thermal cleaning is more effective in reducing the intensities of NRM. Thermal demagnetization curves are of different shapes, from almost square, characteristic of hematite, to steeply decreasing or even increasing curves. In spite of great scatter of intensity and direction (Fig. 2) many samples seem to possess stable directions, usually with blocking temperature above 575°C. Previously published (Løvlie *et al.* 1984) Zijderveld diagrams and directional changes demonstrate quite well defined, stable directions, but very scattered. This scatter may be caused by inadequate vector analysis, which is especially difficult because of the low intensity of magnetization, (c. 10^{-3} Am^{-1}) the multicomponent nature of NRM and overlapping coercivity spectra of particular components.

To uncover particular directions of magnetization vector subtraction has been performed after each demagnetization step and the removed vectors have been fitted to lines by means

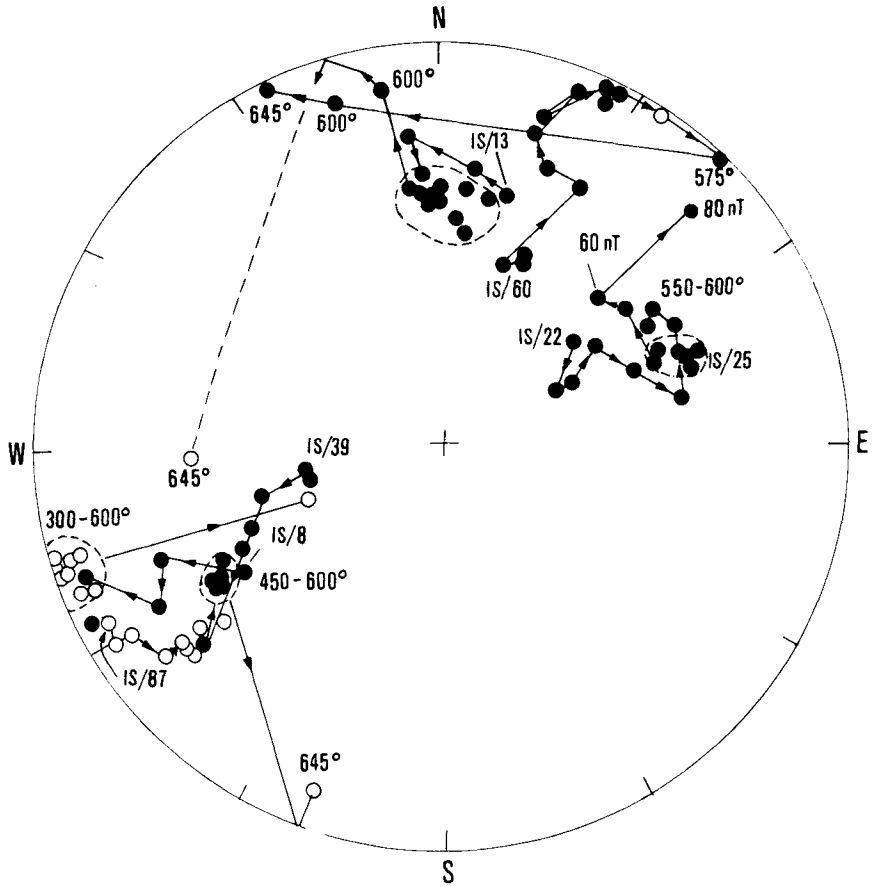


Figure 2. Stereographic projection of directional changes of NRM; full circles – lower hemisphere; open circles – upper hemisphere.

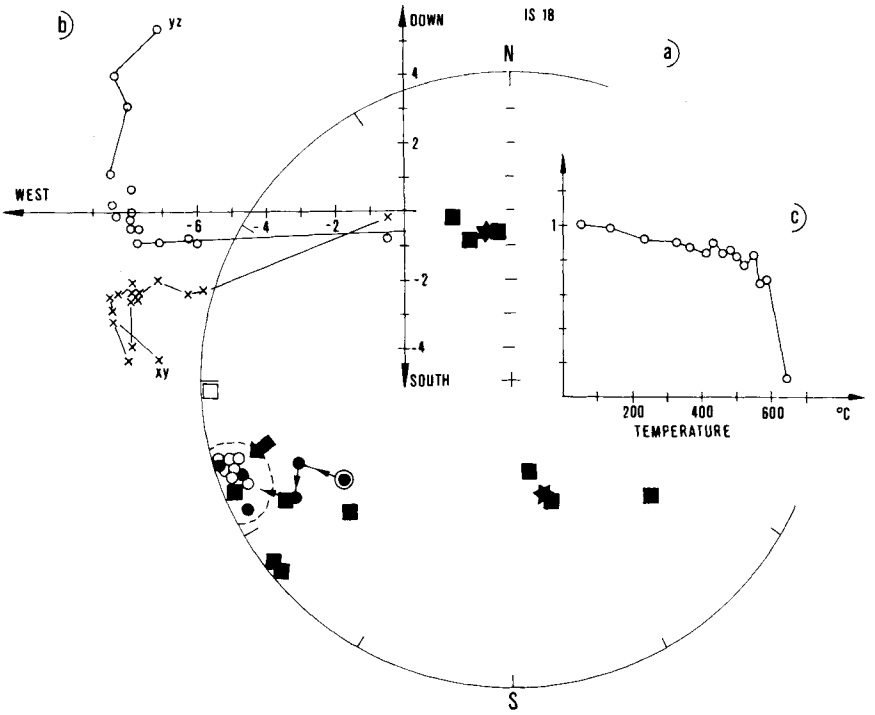


Figure 3. The illustration of an analysis of step demagnetization data; (a) The analysis of demagnetization data demonstrated on stereonet; circles – directional changes of NRM; squares – vectors removed on each step of demagnetization; stars – mean of subtract vector; (b) Zijderveld diagram; circles – *YZ* plane; crosses – *XY* plane. Units on the vector axis 8×10^{-5} Am^{-1}; (c) Demagnetization curve.

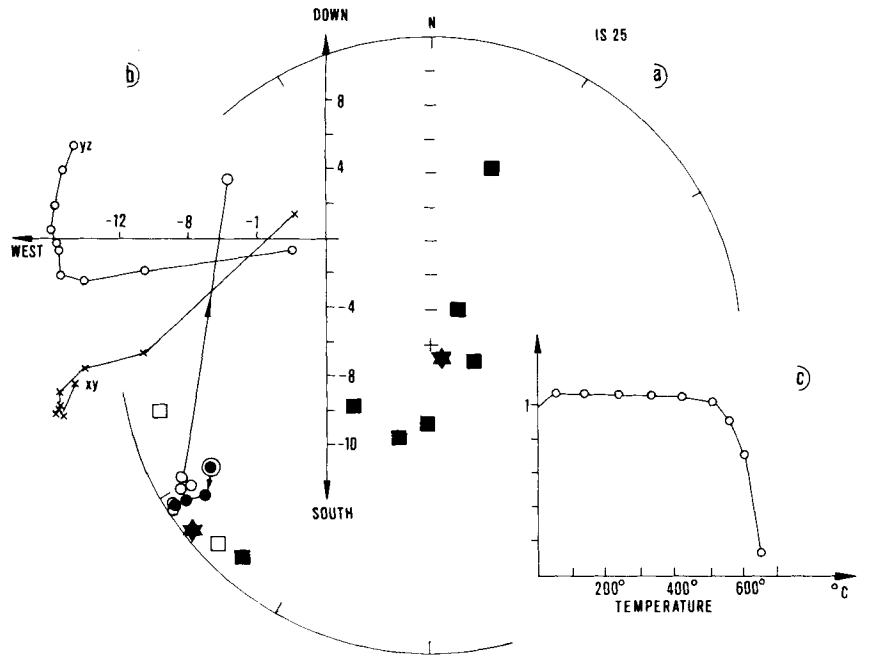


Figure 4. Legends as in Fig. 3.

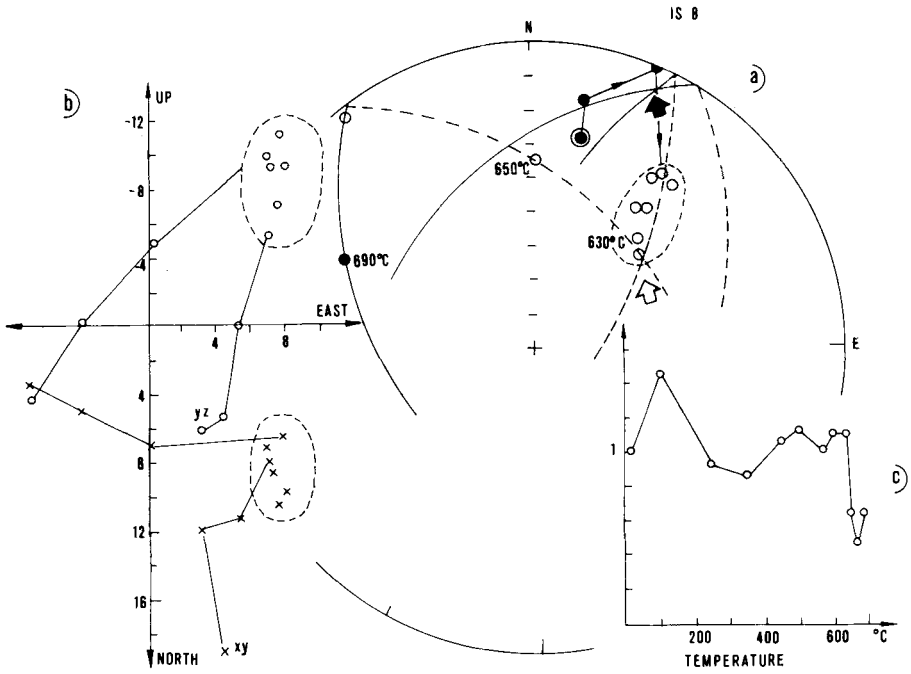


Figure 5. The illustration of an analysis of step demagnetization data. (a) The analysis of demagnetization planes represented by great circles. Arrows indicate the intersection of planes. Further legend as in Fig. 3.

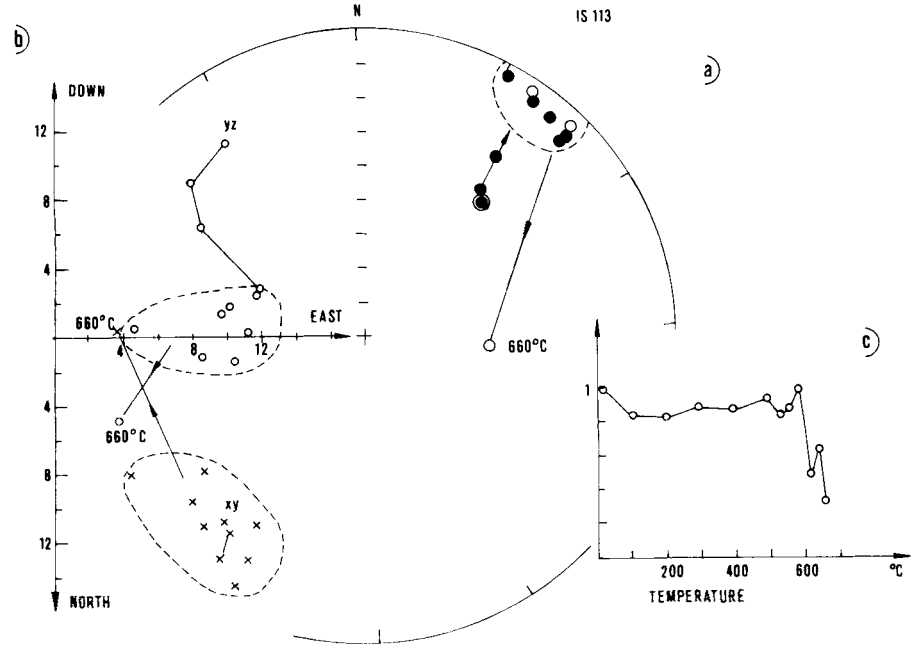


Figure 6. An example of demagnetization data which cannot be analysed by analytical method. Legend as in Fig. 3.

of the least squares method. Next step is to determine a plane in which the removed vectors are confined. If more than one such plane are found, intersection points represent a common direction for both planes (Hoffman & Day 1978; Kirschvink 1980; Jeleńska 1985). The analysis is illustrated in Figs 3, 4, 5 and 6. Fig. 3 shows a simple case of stable remanence which is destroyed at the end of demagnetization treatment. It is possible to evaluate the final direction from the Zijderveld diagram as well as from the directional trend on a stereonet. However, the vector subtraction analytical method gives one vector removed between 200°C and 575°C (stars in Fig. 3) and another at the end of thermal treatment. This agrees well with the direction obtained from the Zijderveld diagram and lies among the cluster of directions on the stereonet. In Fig. 3 the most stable calculated direction is indicated by an arrow. Fig. 4 demonstrates another example of analysis of the demagnetization path, where the analytically calculated vector agrees well with the value estimated from the Zijderveld diagram, but differs slightly from the end-point directions shown on the stereonet. Fig. 5 demonstrates the intersection of demagnetization planes. In some cases it is possible to evaluate the final component by calculating the mean from several directions at the end of a demagnetization path (procedure applied by Roy, Anderson & Lapointe (1979)), whereas neither the Zijderveld diagram nor analytical method enable us to evaluate the characteristic direction. Fig. 6 shows an example of such a case.

Palaeomagnetic results

The vector subtraction analytical method separates shallow directions of remanence in the southwest and northeast quadrants and some directions of steep inclination (Fig. 7). The shallow directions break into groups of normal and reversed polarity. The steep directions fall into three groups (A, B, C). A and B groups are of the same declinations as corresponding shallow directions, and could be the resultant of palaeofield and recent field action. The C-directions are down dipping and in the northwest quadrant. The mean $D = 320°$; $I = 46°$ is close to the palaeocurrent direction in Dicksonfjorden (Friend & Moody-Stuart 1972), and almost perpendicular to the mean axis of maximum susceptibility (Løvlie *et al.* 1984). Square or hexagonal flakes of hematite occur in Dicksonfjorden sandstones in two size grain ranges of about 15 μm and 50 μm. They consist of individual grains or hematite coating on larger matrix granules (Løvlie & Torsvik 1984). As the palaeocurrents may affect the alignment of hematite grains to different extents, depending on their size and shape distribution considering that the magnetic anisotropy of hematite is of magneto-crystalline or stress origin, and the shape of grains is not elongated, palaeocurrents can only be a randomizing agent. High inclination of the C-direction is more evidence against the aligning ability of palaeocurrents.

The C-direction occurred predominantly in samples which carried the remanence of normal polarity, although they do not differ mineralogically from the others. So the ambient magnetic field condition during or shortly after deposition may explain the origin of the C-direction. For example, the C-directions could be the superposition of normal and reversed shallow directions.

The distribution of blocking temperatures is shown in Fig. 8. The shallow directions are of high blocking temperature ranging from 500°C to 700°C (Fig. 8a). The steep directions are associated with lower blocking temperature, except the C-directions which unblocked over a wide range of temperature. The blocking temperature distributions of the shallow and steep directions are shown in Figs. 8(b) and (c).

Figs. 7 and 8 demonstrate the directions separated by the analytical method. To calculate

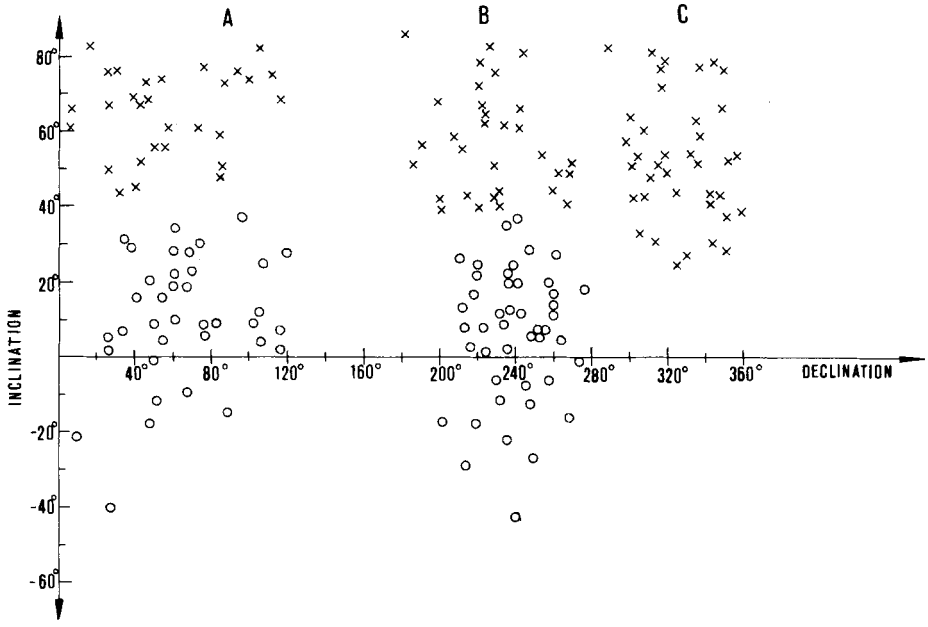


Figure 7. Distribution of different components of NRM presented in orthogonal system. Circles – shallow directions; crosses – steep directions. Letters: A, B, C refers to the steep directions only.

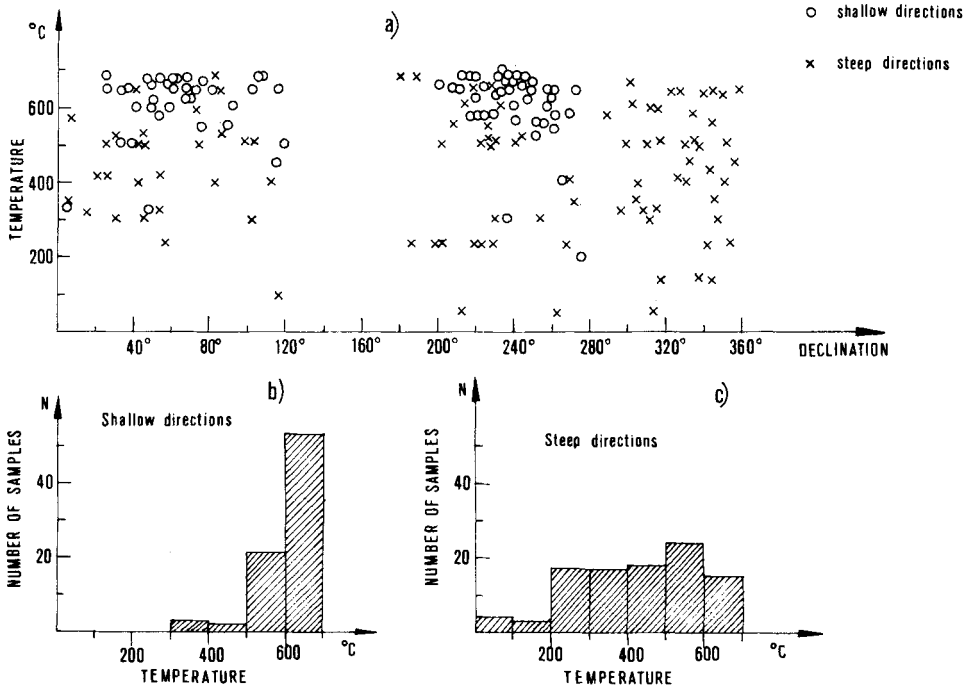


Figure 8. Distribution of blocking temperatures of shallow and steep directions. (a) Distribution in declination plane (see Fig. 7). (b) and (c) Blocking temperature spectra for shallow and deep directions, respectively.

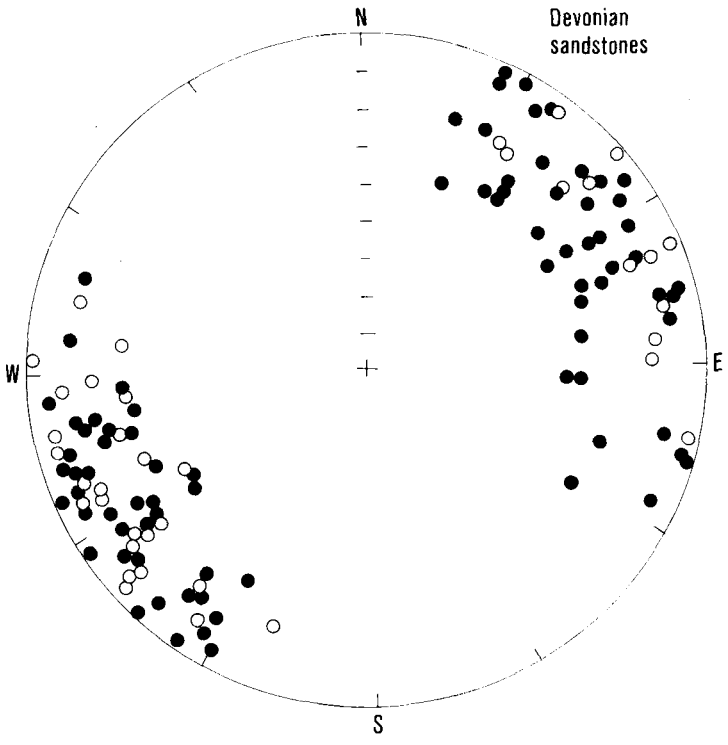


Figure 9. Distribution of shallow directions on stereonet.

further, the directions from the end of the demagnetization path were also taken (the case demonstrated in Fig. 6). All shallow directions are plotted on stereonet in Fig. 9.

The mean site directions before and after bedding correction are listed in Table 1. The site directions were grouped, according to stratigraphic position (Fig. 11). The values of mean directions for groups of samples are listed in Table 2 and its distribution on stereonet is demonstrated in Fig. 10. As the Devonian sandstones lie almost horizontally, the bedding correction does not change significantly the directions.

Local tectonic implications

The basal sequence of Devonian red sandstones in Dicksonfjorden is not exposed. The stratigraphic positions of the sampling sites and groups have been fixed relative to the sequence of Carboniferous limestones overlying the Devonian red bed succession. In a previous paper (Løvlie *et al.* 1984) a correlation between directional distributions for five groups of samples, made by subjective pattern recognition, led to the conclusion that the Dicksonfjorden represent a topographic feature associated with a tectonic structure.

The palaeomagnetic directions presented in this study allow more precise conclusions (Fig. 11). The oldest groups 6, 7 and 3 were deposited in a reversed polarity palaeofield. Succeeding a change in polarity, group 5 (mixed with reversed directions prevailed) and group 4 (mixed with normal directions prevailed) were deposited. The youngest groups (2 and 1) carry the normal polarity magnetization. Note that the direction of group 4 lies more to the east than the others. Such a sequence of groups of samples gives a logical picture of polarity reversal. The tectonic conclusion may now be considered on firmer evidence.

Table 1. Mean directions for sites.

Group	Site	N	D	I	D_F	I_F	k	α_{95}	P
1	2	3	55.4	27.7	54.1	20.2	81.0	13.8	*N*
	19 and 20	11	52.8	21.7	53.3	19.0	9.2	15.9	*N*
	21	13	53.3	26.7	53.9	24.0	13.4	11.7	*N*
2	1	6	42.9	18.5	38.7	23.6	9.5	22.7	*N*
	4	3	52.9	16.3	51.8	7.8	17.3	30.2	*N*
3	3	13	243.6	−7.9	243.5	6.0	26.2	8.2	*R*
	29	5	244.3	−35.8	240.0	−26.1	14.3	20.9	*R*
	30 *R*	4	136.8	−26.3	141.7	−21.5	9.6	31.2	*R*
4	5	5	96.7	13.0	95.5	5.9	14.2	20.9	*N*
	6	7	72.1	19.4	71.2	10.0	7.2	24.0	*M*
	7 and 8	5	78.7	−3.1	79.3	−11.9	5.7	34.5	*M*
5	9	4	85.1	0.6	85.5	−7.7	44.2	13.9	*N*
	10 and 11	9	233.9	−14.4	233.8	−4.4	12.7	15.0	*M*
	12 and 13	9	240.8	−10.0	240.3	−2.4	8.2	19.1	*R*
6	14 and 15	7	249.1	4.5	249.4	7.9	16.1	15.4	*R*
	16 and 17	9	230.2	−8.5	230.1	−4.6	13.8	14.3	*R*
7	18	6	251.2	−3.3	251.1	0.1	5.9	29.8	*R*
	22	3	251.2	8.0	251.1	11.0	9.6	41.3	*R*
	23	6	233.2	2.4	233.1	5.1	5.0	32.8	*R*
Secondary vector *C*			319.7	46.2	329.2	44.2	14.0	7.0	*N*

N – number of specimens; D, I – declination and inclination after tilt correction; D_F, I_F - declination and inclination before tilt correction; k – precision parameter; α_{95} – confidence angle at the 95 per cent probability level. R – direction not used in computation of mean group and overall mean directions; P – polarity.

Table 2. Mean directions for groups.

Group	N	P	D	I	k	α_{95}	D_F	I_F	K	α_{95}
1	27	*N*	52.6	25.9	13.4	7.9	53.6	20.3	10.6	8.8
2	9	*N*	46.4	17.8	12.0	15.5	43.4	18.2	10.6	16.6
3	18	*R*	243.8	−15.5	14.9	9.3	242.6	−2.6	13.3	9.8
4	17	*M*(+)	81.7	11.2	8.1	14.4	81.1	2.6	8.1	14.4
5	22	*M*(−)	242.7	−10.2	9.7	10.5	243.2	−3.5	8.9	10.7
6	16	*R*	238.5	−2.8	12.1	11.1	238.6	0.9	12.0	11.1
7	15	*R*	244.2	1.4	6.2	16.6	244.0	4.4	6.2	16.6

Group	N	P	D	I	k	α_{95}	λ'	ϕ'	δ_m	δ_p
Normal direction	5	*N*	51.4	22.2	144.1	6.4	18.6	141.1	6.8	3.6
Reversed direction	10	*R*	244.3	−6.6	25.2	9.8	8.2	129.7	9.9	5.0
Mean of sites	18	*M*	63.8	11.7	19.9	7.9	10.9	129.6		
Mean of sites excluding group 4	15	*M*	60.1	11.9	23.9	8.0	11.7	133.3	8.1	4.1
Mean of groups	7	*M*	61.5	12.3	34.9	10.3	11.6	131.9		
Mean of groups excluding group 4	6	*M*	58.2	11.9	44.0	10.2	12.0	135.2	10.4	5.3

λ', ϕ', are the latitude and longitude of palaeomagnetic pole in the northern hemisphere, δ_m, δ_p are semi-axes of the oval of confidence of the pole. The other notations as in Table 1. The sign in the parentheses shows which polarity prevails (+: N; − : R).

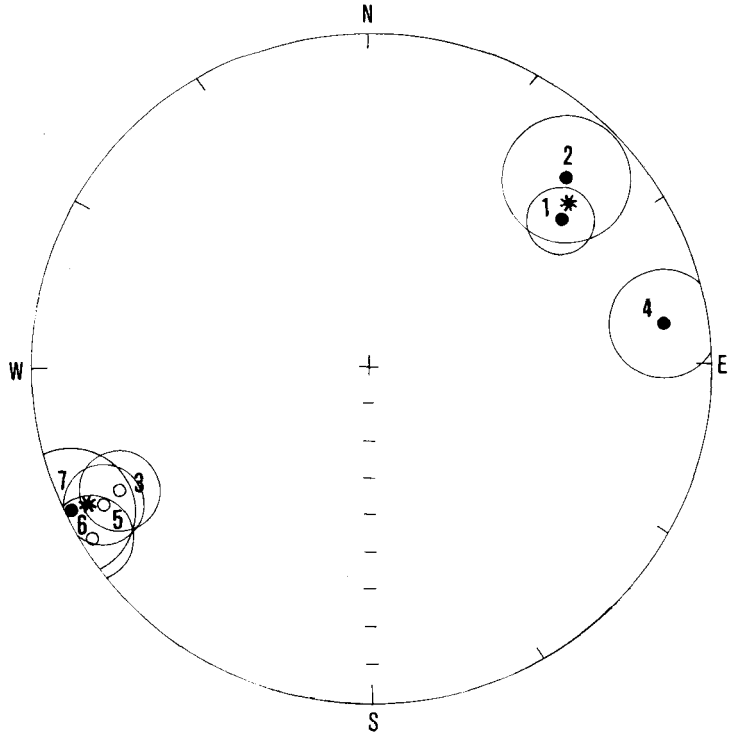


Figure 10. Mean directions of groups shown with the circles of 95 per cent confidence.

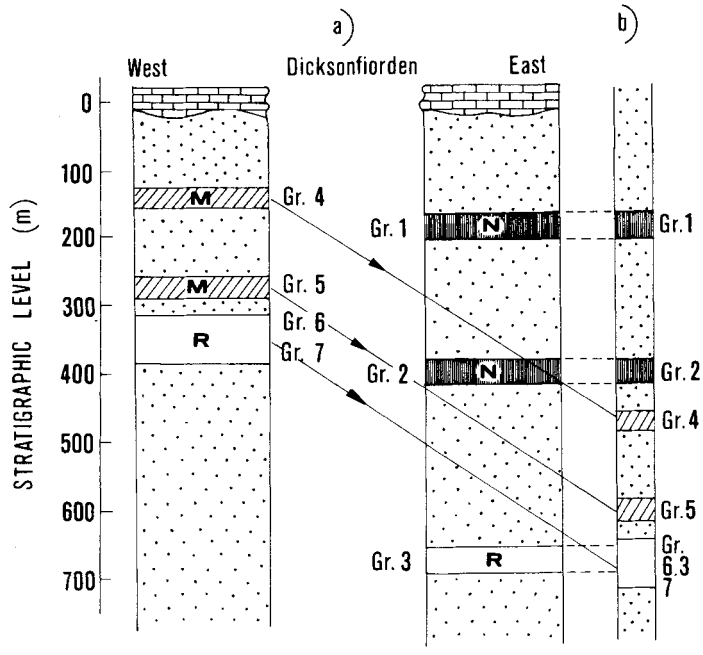


Figure 11. Stratigraphic and polarity correlations between seven sampling areas (groups) on both sides of Dicksonfjorden. Stratigraphic levels relative to base of overlying Middle Carboniferous Limestones; (a) Distribution of sampling areas along stratigraphic column; (b) Polarity sequence after upwards displacement of eastern edge of Dicksonfjorden. N – normal, R – reversed, M – mixed polarity.

Placing the group 3 on the same stratigraphic level as the groups 6 and 7 resulted in a vertical upward displacement of the eastern site of Dicksonfjorden by about 350 m and is associated with a normal fault down-throwing to the east. The proposed fault runs along Dicksonfjorden in a N–S direction and may be related to the Balliolbreen Fault Zone (BFZ), about 20 km to the east.

According to Harland (1978), Harland & Wright (1979) the BFZ was developed in the Upper Devonian as a component of transcurrent sinistral strike-slip movements (the Svalbardian phase). The Cainozoic movements slightly reactivated the BFZ, but were mainly concentrated along West Spitsbergen Orogenic Belt. The Dicksonfjorden fault might have been developed together with the BFZ or just after the Svalbardian phase as adjustment at the end of this activity. It could also be renewed at the time of the West Spitsbergen Orogeny.

The polarity sequence for the groups of samples assuming no displacement is shown as alternative in Fig. 11.

Discussion

Although the within group scatter is rather large, the mean group directions of the Wood Bay Formation seem to be consistent (Table 2 and Fig. 10). The F-distribution tests (Table 3) were applied to the different combinations of groups (McElhinny 1973; Collinson, Creer & Runcorn 1967). The F-distribution test of normal groups 1 and 2 is not significant, showing that these two units belong to the same population. The same applies to groups 6 and 7 of reversed polarity. The reversed groups 5 and 3 also belong to the same population as groups 6 and 7.

The mean palaeomagnetic poles were calculated for reversed, normal, as well as for all directions together. Group 4 was deposited in a time of polarity reversal and carries magnetization of mixed polarity and intermediate direction. Usually, such directions are not

Table 3. *F*-distribution test.

Group	N	K	R			
1	27	13.4	25.0631			
2	9	12.0	8.3321			
3	18	14.9	16.8563			
5	22	9.7	19.8348			
6	16	12.1	14.7570			
7	15	6.2	12.7566			
	N_i		R_T	K_1/K_2	F	Significant
1, 2	36		33.3023	1.12	1.21	no at 90 per cent
6, 7	31		27.4617	1.95	0.43	no at 90 per cent
6, 5	38		34.5002	1.25	0.97	no at 90 per cent
6, 7, 5	53		47.1409	1.95	0.92	no at 90 per cent
5, 3	40		36.6507	1.54	0.46	no at 90 per cent
6, 3	34		31.3897	1.23	3.00	no at 95 per cent
5, 6, 3	56		51.2204	1.54	1.33	no at 90 per cent
3, 7	33		29.2999	2.40	2.62	no at 95 per cent

K – precision parameter; $F = \frac{2(\Sigma N_i - p)}{2(p-1)} \cdot \frac{\Sigma R_i - R_T}{\Sigma N_i - \Sigma R_i}$, where p – number of population being compared; ΣN_i – total number of specimens in p populations; ΣR_i – algebraic sum of resultant vectors R_i of a population; R_T – vector sum of R (Collinson *et al.* 1967).

used to calculate a palaeomagnetic pole. For that reason the pole position calculated excluding group 4 was considered in further discussion.

The palaeomagnetic pole obtained for the sandstones of Wood Bay Formation (Lower Devonian) could not be compared with the other palaeomagnetic poles for Spitsbergen, because up to now the investigations of Devonian sediments in Spitsbergen have been unsuccessful (Storetvedt 1972; Jeleńska, Kruczyk & Kadzialko-Hofmokl 1979).

To find a reference frame to which the Spitsbergen pole might be related, Lower Palaeozoic palaeomagnetic poles for different parts of Europe have been examined, i.e. Ordovician, Silurian and Devonian poles for the British Isles, for the Russian Platform and for continental western Europe (Table 4 and Fig. 12).

Table 4. Palaeozoic poles from the British Isles in comparison with other European regions.

No	Rock unit	Palaeomagnetic pole	Age (Myr) or α_{95}
ORDOVICIAN			
British Isles			
1	Younger Gabbros Aberdeenshire[1]	189E, 11N	486–461
2	Ballantrae Complex, Serpentinites[1]	207E, 12N	475
3	Byne Hill Gabbro[1]	206E, 10N	
4	Foyers Complex[1]	168E, 16N	420–460
5	Trefgarn Andesitic Series[1]	172E, 5N	
6	Eycott Group Lavas[1]	176E, 9N	
7	Stapley Hill Volcanic Group[1]	190E, 20N	
8	Builth Volcanic Series[1]	179E, 16N	
9	Tuffs, Welsh Borderlands[1]	178E, 23N	451
10	Dolerite Intrusions, Builth Inliner[1]	182E, 25N	
	Mean pole (poles 1–10)	185E, 12N	α_{95} = 9.0
	Russian platform mean pole[3]	155E, 34N	α_{95} = 12.0
	Sulitjelma gabbro northern Norway[1]	180E, 14N	
	Ordovician, Holy Cros Mts, Poland[7]	142E, 33N	
SILURIAN			
British Isles			
11	Andesite Intrusions, Shelve Inlier[2]	244E, 14S	
12	Corndon Hill Phacolith[2]	249E, 15S	
13	Dolerite Intrusions, Shelve Inlier[2]	243E, 15S	
14	E–W Dolerite Dykes, Shelve, Longmynd[2]	230E, 16S	
15	Moel-y-Golfa Andesite[2]	242E, 16S	
	Mean pole (poles 11–15)	240E, 15S	α_{95} = 7.1
16	Helmsdale granite[5]	175E, 31N	400–420
17	Strontian granite[4]	154E, 21N	
18	Foyers granite[4]	167E, 27N	
19	Peterhead granite[6]	178E, 21N	
	Mean pole (poles 16–19)	171E, 25N	α_{95} = 8.8
	Russian platform mean pole[3]	159E, 34N	α_{95} = 11.0
	Ringerike Sandstone, Norway[1]	159E, 21N	Su
UPPER SILURIAN			
British Isles			
20	Lavas, Somerset and Gloucestershire[2]	125E, 7S	
21	Lorne Plateau Lavas[2]	141E, 2S	421
22	Garabal Hill-Glen Fyne Intrusion[2]	146E, 5N	404
23	Arrochar Complex[2]	144E, 8S	418
	Mean pole (poles 20–23)	139E, 3S	α_{95} = 12.8

Table 4 – *continued*

No	Rock unit	Palaeomagnetic pole	Age (Myr) or α_{95}
LOWER DEVONIAN			
British Isles			
24	Lower O.R.S. Lavas and sediments Midland Valley[1]	140E, 4N	394
25	Midland Valley Lavas[4]	138E, 2S	
26	Cheviot Hill lavas/granites[4]	149E, 1S	
27	Lower O.R.S. lavas, Cheviot[1]	140E, 11N	390
28	Lower O.R.S. lavas, Glencoe[1]	149E, 6S	
29	Cautley Fell Dolerite[1]	127E, 9N	
30	Fishguard Volcanic Group[1]	119E, 2N	
	Mean pole (poles 24–30)	137E, 2N	α_{95} = 9.4
	Russian platform mean pole[3]	163E, 39N	α_{95} = 3.3
MIDDLE–UPPER DEVONIAN			
British Isles			
31	Shap Granite[1]	133E, 33N	393
32	SW–NE Minetts Dykes, Cross Fell[1]	136E, 23N	
33	Middle O.R.S. Sediments, Foyers[1]	147E, 30N	
34	Hendre Dolerite, Berwyns[1]	166E, 32N	
35	Blodwell Keratophyre, Berwyns[1]	169E, 33N	
36	Upper O.R.S. Jedburgh[1]	169E, 23N	
37	Upper O.R.S. Bristol District[1]	158E, 32N	
38	Lavas, Orkney[1]	150E, 24N	370
39	Caithness Sandstone[1]	147E, 21N	
40	Hoy Lavas[1]	146E, 23N	
41	Duncansby Neck[1]	149E, 24N	
42	John O'Groats sediments[1]	145E, 24N	
43	Shetland ignimbrites[6]	135E, 20N	
	Mean pole (poles 31–43)	150E, 27N	α_{95} = 6.1
	Russian platform mean pole[3]	161E, 32N	α_{95} = 3.2
Continental Europe			
44	Kvamshesten O.R.S., Norway[1]	170E, 22N	*Dm*
45	Röragen Sandstone, Norway[1]	160E, 19N	*DL*
46	Ytteroy dyke Mixed, Norway[1]	140E, 10N	
47	Eifel Sandstone, West Germany[1]	163E, 42N	
48	Diabases East Germany[1]	142E, 35N	
49	Bornholm, Denmark[8]	156E, 14N	
50	Jotnian Upper Dala, Sweden[1]	170E, 26N	
	Mean pole (poles 44–50)	158E, 25N	α_{95} = 10.4

1 Irving, Tanczyk & Hastie (1976); Hicken *et al.* (1972); McElhinny & Cowley (1977, 1980).
2 References as in 1 and Piper (1979).
3 References as in 1 and Khramov (1982).
4 Torsvik (1984).
5 Torsvik *et al.* (1983).
6 Torsvik (1985).
7 Malkowski & Tomczyk (1973).
8 Abrahamsen (1977).

The Ordovician poles are mainly from the British Islands, with only two poles for continental Europe: from Norway and Poland. The Norwegian pole is very dubious, because of scatter of directions (α_{95} = 28°), and is, therefore, shown in Fig. 12 with a question mark.

There is only one Silurian pole for continental Europe, the Upper Silurian Ringerike sandstone from Norway. The British Silurian poles, although numerous, are questionable,

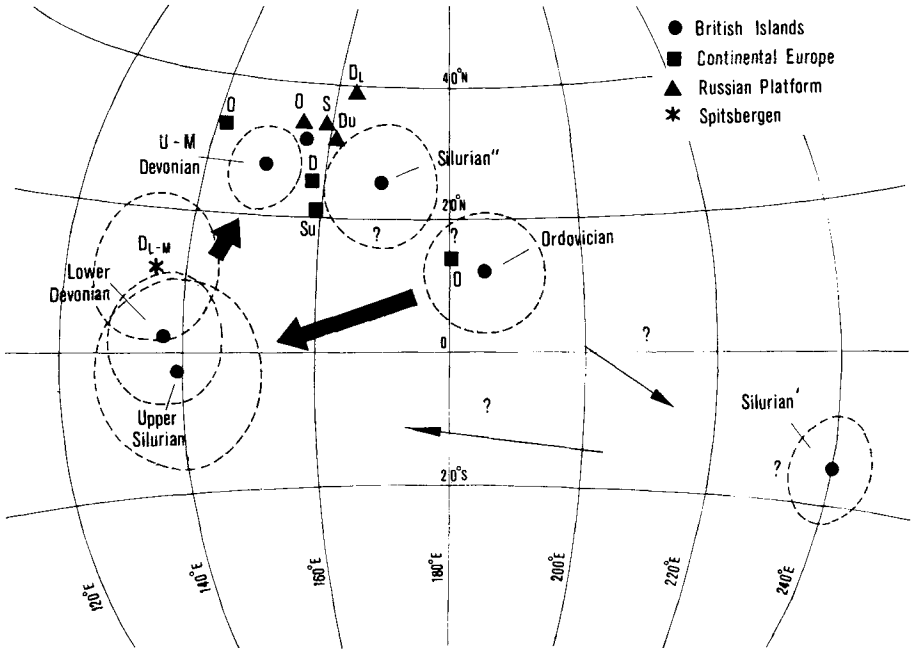


Figure 12. The Lower Palaeozoic poles for the British Isles, Russian Platform and continental Europe together with the Devonian pole for Spitsbergen, O – Ordovician; S – Silurian; D – Devonian. The British and Spitsbergen poles are shown with 95 per cent confidence circle.

because the Middle-Upper Silurian poles marked in Fig. 12 as Silurian′ (Piper 1979) and Silurian″ (Torsvik 1984, 1985) differ significantly. For that reason only Ordovician, Upper Siluran–Lower Devonian and Upper-Middle Devonian fragments of APW curve for British Islands were taken into consideration, leaving the Silurian period for further investigation.

The striking feature of the British APWP is the great mobility of palaeomagnetic pole from Ordovician up to Upper–Middle Devonian time in comparison with the motionless position of poles from the Russian platform and continental, western Europe, for which all Devonian poles lie together. It should be stressed, however, that the single Ordovician and Silurian palaeomagnetic poles and few Devonian data for Europe are insufficient proof for any conclusion about the relative movement between British Islands and Europe, although there are several mobilistic models of Europe (Van der Voo 1982, 1983; Van der Voo & Scotese 1981; Bachtadse, Heller & Kröner 1983) as well as fixistic ones. Piper (1979), in particular, concluded that the British sub-plate was displaced prior to Middle Devonian time and locked between the Baltic Shield and Laurentia as a result of the collision of Gondwana and Laurentia. Opposing this point of view Duff (1980) regarded Europe as a single unit extending from the Baltic Shield to the Armorican Massif and lasting from early Cambrian to early Permian.

The Lower–Middle Devonian pole for Dicksonfjorden Red Bed sediments from Spitsbergen (12°N, 135°E) lies close to the Lower Devonian British poles, indicating that Spitsbergen had its present relationship to the British Isles, so the interpretation of the Spitsbergen palaeomagnetic results depends on which tectonic model of Europe is adopted and this cannot be elucidated without new Lower Palaeozoic data for continental Europe. It should be mentioned that the Permian and Mesozoic poles for Spitsbergen (Vincenz &

Jeleénska 1985) indicate that Spitsbergen was rotated after these times and thus give indirect support to the mobilistic model of northern Europe.

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